

**In one line:** SwiftUI derives an **accessibility tree** from the view tree: every Text, Image and control becomes an element, stacks are transparent. Your job is to **shape** that tree — one stop per idea (group, label, value, actions) — and let **text grow** (Dynamic Type) while the layout **reflows** and motion respects user settings.

## The tree

- **accessibilityElement(children:)** — default **.ignore**: new element, children hidden, *you* label it. **.combine**: children merged into one (“not all traits are merged; a default action may become a named action”). **.contain**: children kept, grouped (scope for order and rotors).
- **Describe**: **accessibilityLabel**, **Value** (changing state), **Hint**, **accessibilityAddTraits(.isButton / .isHeader / .isSelected)**, **accessibilityInputLabels** (Voice Control names), **accessibilityIdentifier** (tests). Decorative: **Image(decorative:)** or **.accessibilityHidden(true)**.
- **Actions**: **.accessibilityAction(named: "Favorite")** { } = the Actions rotor (swipe up/down) — the VoiceOver way to reach swipe / long-press features. Kinds: **.magicTap** (two-finger double tap: play/pause, answer), **.escape** (Z-scrub: close); **accessibilityAdjustableAction** for increment/decrement.
- **Custom controls**: **accessibilityRepresentation** { **Slider(...)** } (15) — SwiftUI hides the stand-in and takes its elements, traits and actions. A custom **ToggleStyle** keeps **Toggle**'s semantics for free.
- **Navigation**: **accessibilityRotor("Unread")** { ... **AccessibilityRotorEntry** } — jump lists; **accessibilitySortPriority(n)** — higher first, default 0; **@AccessibilityFocusState** moves focus (to an error). **accessibilityCustomContent**: secondary facts on demand; **accessibilityChildren** { }: synthetic elements for a Canvas.

## Dynamic Type & settings

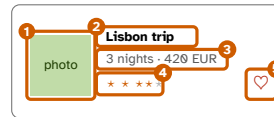
- **12 sizes**: **xSmall ... large** (default) ... **xxxLarge** + **accessibility1-5**; **isAccessibilitySize**. **11 text styles** (**largeTitle ... caption2**); **.body** is 17 pt at large. Custom font: **.custom("Inter", size: 17, relativeTo: .body)**.
- **@ScaledMetric(relativeTo: .body) var s = 24** (14) scales *numbers*: icon size, padding, spacing.
- **Clamp** with **.dynamicTypeSize(...DynamicTypeSize.xxxLarge)** (15) only for chrome that cannot grow — and add **.accessibilityShowsLargeContentViewer()**.
- **Reflow**: **ViewThatFits { HStack; VStack }** (16) or **isAccessibilitySize ? AnyLayout(VStackLayout()) : AnyLayout(HStackLayout())** (keeps identity); no fixed heights, let text wrap, allow scrolling.
- **Settings** (environment): **accessibilityReduceMotion** (fade instead of move), **accessibilityReduceTransparency**, **colorSchemeContrast**, **accessibilityDifferentiateWithoutColor**, **legibilityWeight**, **accessibilityVoiceOverEnabled**.

## Test it

|                                          |                                                                                                      |
|------------------------------------------|------------------------------------------------------------------------------------------------------|
| Accessibility Inspector                  | inspect elements, run an audit, flip settings live                                                   |
| Xcode previews                           | Dynamic Type / colour-scheme variants side by side                                                   |
| <code>performAccessibilityAudit()</code> | XCUITest, iOS 17; 9 types incl. contrast, dynamicType, textClipped, hitRegion; handler true = ignore |
| <code>voiceOverService</code>            | iOS 27 XCUITest: <code>moveForward()</code> , <code>currentSpeech()</code>                           |
| real VoiceOver                           | triple-click shortcut; listen to order and wording                                                   |

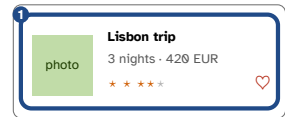
## What VoiceOver visits

default: 5 stops



"photo" · "Lisbon trip" · "3 nights..." · "star, star, ..." · "heart, button" — five swipes, no meaning of the whole

**.ignore + label/value/action**



"Lisbon trip, 3 nights, 420 euros. 4 of 5 stars. Button." Swipe up/down: **Favorite**

label = *what* (title + subtitle) · value = *state* (rating) · trait = *role* (button) · named action = the heart. **.combine** merges what the children *already* say — decide the wording yourself with **.ignore**.

## Example — one card, one stop

```
struct TripCard: View {
    let trip: Trip; @Binding var fav: Bool
    @ScaledMetric(relativeTo: .body) private var thumb = 56
    @Environment(\.dynamicTypeSize) private var size
    var body: some View {
        let layout = size.isAccessibilitySize // reflow, keep identity
        ? AnyLayout(VStackLayout()) : AnyLayout(HStackLayout())
        layout {
            Photo(trip).frame(width: thumb, height: thumb)
            Details(trip); FavButton(isOn: $fav) // stars, heart
            .accessibilityElement(children: .ignore) // one stop
            .accessibilityLabel("\(trip.title), \(trip.subtitle)")
            .accessibilityValue("\(trip.rating) of 5 stars")
            .accessibilityAddTraits(.isButton)
            .accessibilityAction(named: fav ? "Unfavorite" : "Favorite") {
                fav.toggle()
            }
        }
    }
}
// UI test: try XCUIApplication().performAccessibilityAudit()
```

## Traps

- ! **.accessibilityElement()** with no argument = **.ignore**: children vanish — an unlabelled, silent element.
- ! **.onTapGesture** card: no **.isButton** trait — VoiceOver users can't tell it acts.
- ! **.frame(height: 44) / lineLimit(1)** on text — clipped at AX sizes (**textClipped**).
- ! Clamping Dynamic Type app-wide to dodge layout work; state shown only by colour or motion.
- ! A green audit ≠ accessible: it can't judge order, wording or task flow — listen with VoiceOver.

## Remember

**“One stop per idea: group, label, value, action. Text grows, layout reflows, motion calms.”**